By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

Macaulay Beam Analysis - BetaTesting

11 - Steel - Single Span - Settlement

Bridge, L =		in		E =	29,000,000 lb / in^2	Length Units	in
Beam Sect =	W14x61	W14x90		G =	1.740E+06 lb / in^2 = 0.06*29,000,000	Force Units	lb
A =	17.90	26.50	in^2				
I =	640.00	999.00	in^4				

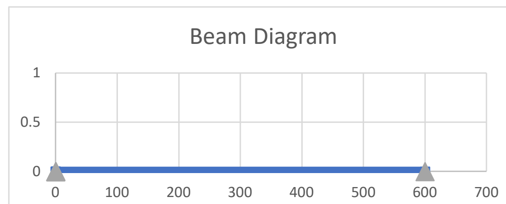
Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in^2	in^4	lb / in^2	lb / in^2
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

	Supp. No.	X	Imposed Movement * dY (vert)	Imposed Movement ** dr (rot)
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	-2.0	0.000
	3			
	4			
	5			

* (settlement) ** (Not Working)

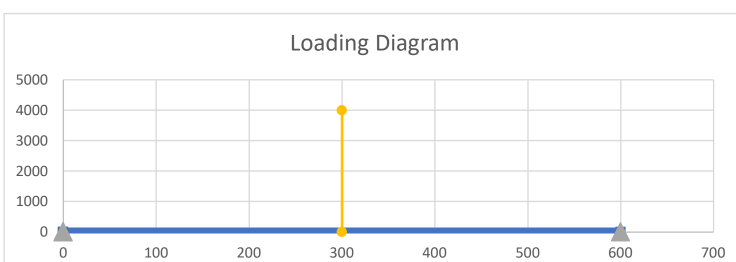


Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)	Load
		in	in	lb / in	lb / in	lb / ft
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0	0.0
	5					

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	300	-4000.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			



By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

BEAM REACTIONS

Position (X)	nforced Disp (dy)	orced Rot (theta)	Vert Reac (Ry)	loment Ract (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	2,000.0	0.0	Support reaction
600.00	-2.00	0.00	2,000.0	0.0	Support reaction

Vert Reac
(kip)
2.000
2.000

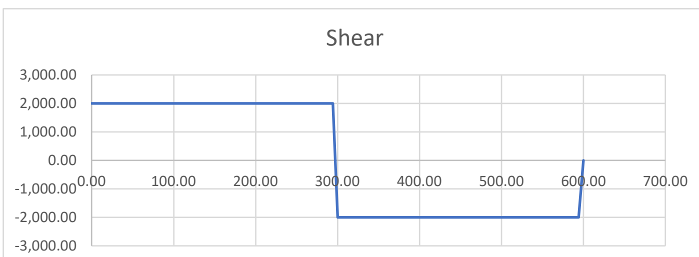
BEAM REACTIONS EQUILIBRUM CHECK

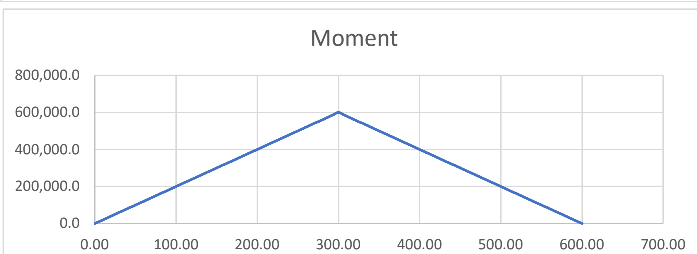
Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-4,000.0	4,000.0	0.0	OK
Sum M @ x=0	F-L	-1.2000E+06	1.2000E+06	0.0	OK

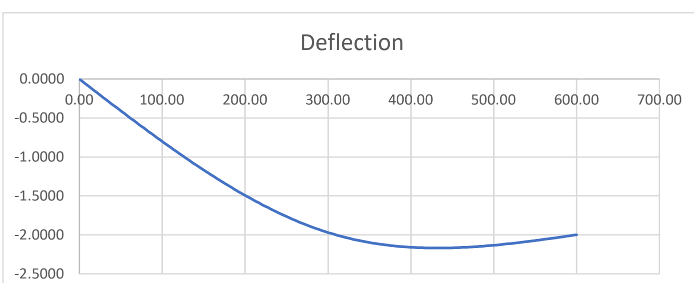
BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	0.00	2,000.0	0.0	0.0000
Max -V	300.00	-2,000.0	600,000.0	-1.9697
Max +M	300.00	-2,000.0	600,000.0	-1.9697
Max -M	0.00	2,000.0	0.0	0.0000
Max Defl	432.00	-2,000.0	336,000.0	-2.1694

Max/Min Value	
(---)	
2.00	kip
-2.00	kip
50.00	kip-ft
0.00	kip-ft
-2.169	in

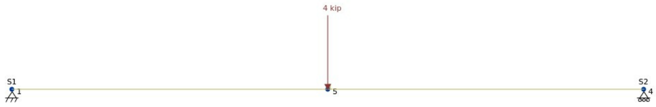






steel-2segment SkyCiv Report

Wed 10 Jun 2026, 02:46PM (GMT-04:00)



File Name: steel-2segment
Software: SkyCiv Structural 3D v7.2.8 (Lic. No.: 1TgQdZH4tY4w7B6tTe9n93aY)
Analysis Type: Linear Static Analysis

Included in this Report:
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Bill of Materials
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SKYCIV FREE TRIAL EDITION

Job Setup

steel-2segment SkyCiv Report
SkyCiv Structural 3D v7.2.8
Licence Number: 1TgQdZH4tY4w7B6tTe9n93aY
Date: Wed 10 Jun 2026, 02:46PM (GMT-04:00)
Analysis Type: LINEAR STATIC ANALYSIS

File Name	steel-2segment
Job Name	Empty
Designer	Empty
Job Description	Empty

Length Units	ft
Section Length Units	in
Force Units	kip
Moment and Torsion Units	kip-ft
Pressure Units	psf
Material Strength Units	psi
Material Density Units	lb/ft^3
Mass Units	kip
Temperature Units	degF
Translation Units	in
Stress Units	psi
Nodes	3
Members	1
Plates	0
Meshed Plates	0
Supports	2
Sections	11
Point Loads	1
Distributed Loads	0
Moments	0
Member Prestress Loads	0
Thermal Loads	0
Pressures	0
Area Loads	0
Self Weight	OFF
User Defined Nodal Masses	0
Auto Defined Nodal Masses	0
Spectral Loads	0
Auto-Stabilize Model	NO
Member Evaluation Points	9
Extrapolate Plate Results From Gauss Points	YES
General Constraint	RRRRRR
Total Degrees of Freedom	12

NODE COORDINATES (ft)

ID	X Coordinate	Y Coordinate	Z Coordinate
1	0.000	0.000	0.000
4	50.000	0.000	0.000
5	25.000	0.000	0.000

MEMBERS (deg, ft, in, ft, kip)

F=Fixed, R=Released, S=Spring

ID	Node A	Node B	Length	Type	Section ID	Rotation Angle	Node A Fixity	Node B Fixity
1	1	4	50.000	Continuous & Normal	10	0.000	FFFFFF	FFFFFF

ID	Node A Offsets	Node B Offsets	Offset Axis	Node A Rot Stiff Y	Node A Rot Stiff Z	Node B Rot Stiff Y	Node B Rot Stiff Z	Cable Length	T/C Limit	Mirror	Disable Non-Linear
1	0, 0, 0	0, 0, 0	Local	-	-	-	-	-	-	No	No

SUPPORTS (kip/ft, kip-ft/rad)

F=Fixed, R=Released, S=Spring

B=Both Axes Restraint, N=Negative-Axis Restraint Only, P=Positive-Axis Restraint Only

ID	Node ID	Restraint Code	Direction Code	X Trans Stiffness	Y Trans Stiffness	Z Trans Stiffness	X Rot Stiffness	Y Rot Stiffness	Z Rot Stiffness	Source
1	1	FFFFFR	BBBBBB	-	-	-	-	-	-	User Defined Nodal Support
2	4	RFFRRR	BBBBBB	-	-	-	-	-	-	User Defined Nodal Support

SETTLEMENTS (in, rad)

ID	Node ID	X Translation	Y Translation	Z Translation	X Rotation	Y Rotation	Z Rotation
1	4	0.000	-2.000	0.000	0.000	0.000	0.000

MATERIALS (psi, lb/ft^3, 10^{-6}/degF)

Shear Modulus is used for members only.
Ex, Ey, Gxy, Gxz, Gyz are used for orthotropic plates only.

ID	Name	Young's Modulus	Shear Modulus	Density	Poisson's Ratio	Thermal Exp. Coeff.	Young's Modulus x	Young's Modulus y	Shear Modulus xy	Shear Modulus xz	Shear Modulus yz	Poisson's Ratio xy
1	American Standard (ASTM) - ASTM A36 - Structural Carbon Steel - Standard	29000000.000	Auto	490.752	0.260	6.670	-	-	-	-	-	Auto

SECTIONS (in, in^2, in^4, deg)

ID	Name	Shape	Material ID	Depth	Width	Shear Area z (STRESS)	Shear Area y (STRESS)	Shear Area z (TIMO)	Shear Area y (TIMO)	Torsion Radius
1	HSS5x5x3/16	HSS5x5x3/16	4	5.000	5.000	1.416	1.416	-	-	3.464
2	W12x14	W12x14	3	11.900	3.970	1.631	2.286	-	-	0.404
3	W14x22	W14x22	3	13.700	5.000	3.022	3.055	-	-	0.557
4	16 x 16	Rectangular	5	16.000	16.000	213.334	213.334	-	-	10.798
5	24 x 16	Rectangular	5	24.000	16.000	320.002	320.000	-	-	13.553
6	3 x 3	Hollow Rectangular	1	3.000	3.000	1.207	1.207	-	-	1.896
7	4 x 4	Rectangular	9	3.500	3.500	10.208	10.208	-	-	2.362
8	4 x 2	Rectangular	9	3.500	1.500	4.375	4.375	-	-	1.437
9	6 x 2	Rectangular	9	5.500	1.500	6.876	6.875	-	-	1.492
10	W14x61	W14x61	1	13.900	10.000	11.277	4.990	-	-	0.991
11	W14x90	W14x90	1	14.000	14.500	17.741	5.702	-	-	1.055

ID	Centroid y	Centroid z	Area (A)	y-Axis MoI (Iy)	z-Axis MoI (Iz)	Torsion Constant (J)	Principal Angle	Non Prismatic
1	2.500	2.500	3.280	12.600	12.600	19.900	0.000	N
2	5.950	1.985	4.160	2.360	88.600	0.070	0.000	N
3	6.850	2.500	6.490	7.000	199.000	0.208	0.000	N
4	8.000	8.000	256.000	5461.333	5461.333	9213.015	0.000	N
5	12.000	8.000	384.000	8192.000	18432.000	19244.519	0.000	N
6	1.500	1.500	2.750	3.495	3.495	5.449	0.000	N
7	1.750	1.750	12.250	12.505	12.505	21.096	0.000	N
8	1.750	0.750	5.250	0.984	5.359	2.875	0.000	N
9	2.750	0.750	8.250	1.547	20.797	5.124	0.000	N
10	6.950	5.000	17.900	107.000	640.000	2.190	0.000	N
11	7.000	7.250	26.500	362.000	999.000	4.060	0.000	N

ID	Area Red. Factor	Iy Red. Factor	Iz Red. Factor	Torsion Constant Red. Factor	Shear Area Y Red. Factor	Shear Area Z Red. Factor
1	-	-	-	-	-	-
2	-	-	-	-	-	-
3	-	-	-	-	-	-
4	-	-	-	-	-	-
5	-	-	-	-	-	-
6	-	-	-	-	-	-
7	-	-	-	-	-	-
8	-	-	-	-	-	-
9	-	-	-	-	-	-
10	-	-	-	-	-	-
11	-	-	-	-	-	-

POINT LOADS (kip)

ID	Load Group	Node	Member	Position (%)	X Magnitude	Y Magnitude	Z Magnitude
1	D1	5			0.000	-4.000	0.000

LOAD COMBINATIONS

ID	Name	Criteria	D1 Factor
1	1: 1.40D1 + 1.40SW1	Strength	1.400
2	2: 1.20D1 + 1.20SW1 + 1.60L1	Strength	1.200
3	4: 1.20D1 + 1.20SW1 + 1.60L1 + 0.30S1	Strength	1.200
4	9: 1.20D1 + 1.20SW1 + 1.00S1	Strength	1.200
5	10: 1.20D1 + 1.20SW1 + 1.00S1 + 1.00L1	Strength	1.200
6	11: 1.20D1 + 1.20SW1 + 1.00S1 + 0.50W1	Strength	1.200
7	15: 1.20D1 + 1.20SW1 + 1.00L1 + 1.00W1	Strength	1.200
8	17: 1.20D1 + 1.20SW1 + 1.00L1 + 1.00W1 + 0.30S1	Strength	1.200
9	23: 1.20D1 + 1.20SW1 + 1.00W1	Strength	1.200
10	28: 0.90D1 + 0.90SW1	Strength	0.900
11	29: 0.90D1 + 0.90SW1 + 1.60L1	Strength	0.900
12	31: 0.90D1 + 0.90SW1 + 1.60L1 + 0.30S1	Strength	0.900
13	36: 0.90D1 + 0.90SW1 + 1.00S1	Strength	0.900
14	37: 0.90D1 + 0.90SW1 + 1.00S1 + 1.00L1	Strength	0.900
15	38: 0.90D1 + 0.90SW1 + 1.00S1 + 0.50W1	Strength	0.900
16	42: 0.90D1 + 0.90SW1 + 1.00L1 + 1.00W1	Strength	0.900
17	44: 0.90D1 + 0.90SW1 + 1.00L1 + 1.00W1 + 0.30S1	Strength	0.900
18	50: 0.90D1 + 0.90SW1 + 1.00W1	Strength	0.900

LOAD CASES

Design Code: ASCE-7-2022-LRFD

Load Group	Design Load Case
D1	D

Bill of Materials

BILL OF MATERIALS FOR MEMBERS (ft, kip)

Section	Material	Quantity	Unit Length	Total Length	Unit Mass	Total Mass
10: W14x61	1: American Standard (ASTM) - ASTM A36 - Structural Carbon Steel - Standard	1	50.000	50.000	3.050	3.050
						3.050

Nodal Results

Load Case: D = D1

NODE REACTIONS (kip, kip-ft)

Support ID	Node	X Force	Y Force	Z Force	X Moment	Y Moment	Z Moment
1	1	0.000	2.000	0.000	0.000	0.000	0.000
2	4	0.000	2.000	0.000	0.000	0.000	0.000
	Reaction Sum	0.000	4.000	0.000			
	Load Sum	0.000	-4.000	0.000			
	Equilibrium	0.000	0.000	0.000			

MEMBER END FORCES AND MOMENTS (kip, kip-ft)

Member	Node	Axial Force	Y Shear	Z Shear	X Torsion	Y Moment	Z Moment
1	1	0.000	2.000	0.000	0.000	0.000	0.000
	4	0.000	-2.000	0.000	0.000	0.000	0.000

NODAL DISPLACEMENTS (in)

Node	X Translation	Y Translation	Z Translation	Total Translation
1	0.000	0.000	0.000	0.000
4	0.000	-2.000	0.000	2.000
5	0.000	-1.970	0.000	1.970

Load Group: D1

NODE REACTIONS (kip, kip-ft)

Support ID	Node	X Force	Y Force	Z Force	X Moment	Y Moment	Z Moment
1	1	0.000	2.000	0.000	0.000	0.000	0.000
2	4	0.000	2.000	0.000	0.000	0.000	0.000
	Reaction Sum	0.000	4.000	0.000			
	Load Sum	0.000	-4.000	0.000			
	Equilibrium	0.000	0.000	0.000			

MEMBER END FORCES AND MOMENTS (kip, kip-ft)

Member	Node	Axial Force	Y Shear	Z Shear	X Torsion	Y Moment	Z Moment
1	1	0.000	2.000	0.000	0.000	0.000	0.000
	4	0.000	-2.000	0.000	0.000	0.000	0.000

NODAL DISPLACEMENTS (in)

Node	X Translation	Y Translation	Z Translation	Total Translation
1	0.000	0.000	0.000	0.000
4	0.000	-2.000	0.000	2.000
5	0.000	-1.970	0.000	1.970

Member Results

Load Case: D = D1

INTERNAL MEMBER FORCES AND MOMENTS (ft, kip, kip-ft)

= Maximum value of a result in the member/plate.
= Minimum value of a result in the member/plate.

Member	Section	Station Location	Axial Force	Y Shear	Z Shear	X Torsion	Y Moment	Z Moment
1	10	0.000	0.000	2.000	0.000	0.000	0.000	0.000
1	10	3.125	0.000	2.000	0.000	0.000	0.000	6.250
1	10	6.250	0.000	2.000	0.000	0.000	0.000	12.500
1	10	9.375	0.000	2.000	0.000	0.000	0.000	18.750
1	10	12.500	0.000	2.000	0.000	0.000	0.000	25.000
1	10	15.625	0.000	2.000	0.000	0.000	0.000	31.250
1	10	18.750	0.000	2.000	0.000	0.000	0.000	37.500
1	10	21.875	0.000	2.000	0.000	0.000	0.000	43.750
1	10	25.000	0.000	-2.000	0.000	0.000	0.000	50.000
1	10	28.125	0.000	-2.000	0.000	0.000	0.000	43.750
1	10	31.250	0.000	-2.000	0.000	0.000	0.000	37.500
1	10	34.375	0.000	-2.000	0.000	0.000	0.000	31.250
1	10	37.500	0.000	-2.000	0.000	0.000	0.000	25.000
1	10	40.625	0.000	-2.000	0.000	0.000	0.000	18.750
1	10	43.750	0.000	-2.000	0.000	0.000	0.000	12.500
1	10	46.875	0.000	-2.000	0.000	0.000	0.000	6.250
1	10	50.000	0.000	-2.000	0.000	0.000	0.000	0.000

MEMBER DISPLACEMENTS (ft, in, rad)

= Maximum value of a result in the member/plate.
= Minimum value of a result in the member/plate.

Member	Station Location	Global X Translation	Global Y Translation	Global Z Translation	Total Translation	Global X Rotation	Global Y Rotation	Global Z Rotation
1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	-0.008
1	3.125	0.000	-0.306	0.000	0.306	0.000	0.000	-0.008
1	6.250	0.000	-0.606	0.000	0.606	0.000	0.000	-0.008
1	9.375	0.000	-0.895	0.000	0.895	0.000	0.000	-0.008
1	12.500	0.000	-1.167	0.000	1.167	0.000	0.000	-0.007
1	15.625	0.000	-1.416	0.000	1.416	0.000	0.000	-0.006
1	18.750	0.000	-1.636	0.000	1.636	0.000	0.000	-0.005
1	21.875	0.000	-1.823	0.000	1.823	0.000	0.000	-0.004
1	25.000	0.000	-1.970	0.000	1.970	0.000	0.000	-0.003
1	28.125	0.000	-2.073	0.000	2.073	0.000	0.000	-0.002
1	31.250	0.000	-2.136	0.000	2.136	0.000	0.000	-0.001
1	34.375	0.000	-2.166	0.000	2.166	0.000	0.000	-0.000
1	37.500	0.000	-2.167	0.000	2.167	0.000	0.000	0.000
1	40.625	0.000	-2.145	0.000	2.145	0.000	0.000	0.001
1	43.750	0.000	-2.106	0.000	2.106	0.000	0.000	0.001
1	46.875	0.000	-2.056	0.000	2.056	0.000	0.000	0.001
1	50.000	0.000	-2.000	0.000	2.000	0.000	0.000	0.002

Load Group: D1

INTERNAL MEMBER FORCES AND MOMENTS (ft, kip, kip-ft)

= Maximum value of a result in the member/plate.
= Minimum value of a result in the member/plate.

Member	Section	Station Location	Axial Force	Y Shear	Z Shear	X Torsion	Y Moment	Z Moment
1	10	0.000	0.000	2.000	0.000	0.000	0.000	0.000
1	10	3.125	0.000	2.000	0.000	0.000	0.000	6.250
1	10	6.250	0.000	2.000	0.000	0.000	0.000	12.500
1	10	9.375	0.000	2.000	0.000	0.000	0.000	18.750
1	10	12.500	0.000	2.000	0.000	0.000	0.000	25.000
1	10	15.625	0.000	2.000	0.000	0.000	0.000	31.250
1	10	18.750	0.000	2.000	0.000	0.000	0.000	37.500
1	10	21.875	0.000	2.000	0.000	0.000	0.000	43.750
1	10	25.000	0.000	-2.000	0.000	0.000	0.000	50.000
1	10	28.125	0.000	-2.000	0.000	0.000	0.000	43.750
1	10	31.250	0.000	-2.000	0.000	0.000	0.000	37.500
1	10	34.375	0.000	-2.000	0.000	0.000	0.000	31.250
1	10	37.500	0.000	-2.000	0.000	0.000	0.000	25.000
1	10	40.625	0.000	-2.000	0.000	0.000	0.000	18.750
1	10	43.750	0.000	-2.000	0.000	0.000	0.000	12.500
1	10	46.875	0.000	-2.000	0.000	0.000	0.000	6.250
1	10	50.000	0.000	-2.000	0.000	0.000	0.000	0.000

MEMBER DISPLACEMENTS (ft, in, rad)

= Maximum value of a result in the member/plate.
= Minimum value of a result in the member/plate.

Member	Station Location	Global X Translation	Global Y Translation	Global Z Translation	Total Translation	Global X Rotation	Global Y Rotation	Global Z Rotation
1	0.000	0.000	0.000	0.000	0.000	0.000	0.000	-0.008
1	3.125	0.000	-0.306	0.000	0.306	0.000	0.000	-0.008
1	6.250	0.000	-0.606	0.000	0.606	0.000	0.000	-0.008
1	9.375	0.000	-0.895	0.000	0.895	0.000	0.000	-0.008
1	12.500	0.000	-1.167	0.000	1.167	0.000	0.000	-0.007
1	15.625	0.000	-1.416	0.000	1.416	0.000	0.000	-0.006
1	18.750	0.000	-1.636	0.000	1.636	0.000	0.000	-0.005
1	21.875	0.000	-1.823	0.000	1.823	0.000	0.000	-0.004
1	25.000	0.000	-1.970	0.000	1.970	0.000	0.000	-0.003
1	28.125	0.000	-2.073	0.000	2.073	0.000	0.000	-0.002
1	31.250	0.000	-2.136	0.000	2.136	0.000	0.000	-0.001
1	34.375	0.000	-2.166	0.000	2.166	0.000	0.000	-0.000
1	37.500	0.000	-2.167	0.000	2.167	0.000	0.000	0.000
1	40.625	0.000	-2.145	0.000	2.145	0.000	0.000	0.001
1	43.750	0.000	-2.106	0.000	2.106	0.000	0.000	0.001
1	46.875	0.000	-2.056	0.000	2.056	0.000	0.000	0.001
1	50.000	0.000	-2.000	0.000	2.000	0.000	0.000	0.002